

A source-pathway-receptor framework for quantifying ecological risk to shallow- and deep-rooted vegetation under heavy metal contamination

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Research Article

Keywords: Heavy metal contamination, Vegetation-based ecological risk, Shallow-root vegetation, Deep-root vegetation

Posted Date: May 20th, 2026

DOI: <https://doi.org/10.21203/rs.3.rs-9187250/v1>

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Additional Declarations: No competing interests reported.

Abstract

Heavy metal contamination in river basins poses a global challenge to vegetation health, crop productivity, and long-term ecological security. Existing vegetation ecological risk assessment methods typically assume uniform vegetation exposure and often overlook the influence of contamination sources and hydrological pathways on vegetation vulnerability. This study develops a vegetation ecological risk assessment framework using a source-pathway-receptor approach to quantify spatiotemporal risk patterns at the root level. An integrated environmental risk assessment was first conducted for the pre- and post-monsoon seasons of 2023. Subsequently, contamination source intensity, environmental pathway intensity, and vegetation receptor vulnerability were combined using the entropy weighting method to estimate vegetation ecological risk within shallow-root (S-VER) and deep-root (D-VER) systems. The derived risk patterns were validated using vegetation response obtained from a normalized difference vegetation index.

Findings revealed a moderate level of S-VER influenced by surface hazards in 49% and 43% of sub-basins during the pre- and post-monsoon season. Seasonal surface hydrological processes regulated these patterns within the ecological function and agricultural production zones. Conversely, D-VER exhibited relatively stable patterns with low-to-moderate risk observed in 51% and 46% of the sub-basins due to persistent subsurface contamination and limited seasonal variability. The lower Hindon Basin exhibits elevated ecological risk due to cumulative upstream contamination loading, intensified hydrological forcing, and reduced vegetation resilience. A negative relationship between vegetation response and estimated ecological risk further supports these patterns. The study provides a scientific basis for evaluating vegetation ecological risk in contaminated river basins and underscores the need for targeted monitoring and sustainable watershed management strategies.

1. Introduction

Vegetations are the primary ecological receptor that regulates the hydrological cycle, enhances soil stability, filters environmental pollutants, and maintains agricultural productivity and long-term ecological balance. However, the health of these ecological receptors is increasingly threatened by heavy metal inputs associated with rapid expansion of small- and large-scale industrial activities, intensified agricultural practices, conversion of forest and barren lands into agricultural and built-up areas, and the construction of artificial structures along river corridors (Duan et al., 2024). In particular, the discharge of untreated industrial effluents, infiltration of contaminated wastewater, and agrochemical runoff have impaired the suitability of irrigation water and soil systems. Once introduced, contaminants enter the soil through seepage and percolation processes. These contaminants are subsequently absorbed by plant roots and transported to plant tissues, where they interfere with normal metabolic functions (Liang et al., 2026; Mishra et al., 2023). The study by Uslu et al. (2025) reported a marked reduction in seed germination when irrigation water enriched with heavy metals was used. In addition, drip and flood irrigation practices can result in direct deposition of heavy metals onto vegetation surfaces. Consequently, across diverse climatic and agricultural regions worldwide, hazardous metals such as cadmium (Cd), lead (Pb), chromium (Cr), copper (Cu), and nickel (Ni) have been identified as detrimental to food crops, tree species, plantation, and grasses, manifesting as chlorosis, stunted growth, reduced biomass, and premature senescence (Alegbe et al., 2025; Rai et al., 2019).

According to the previous studies, ecological risk from heavy metal contamination is shaped by multiple environmental controlling factors such as industrial density, proximity to discharge points, land-use pattern, soil properties, terrain characteristics, and seasonal hydrological variability (Zahid et al., 2025; Das and Behera, 2024; Chen et al., 2024; Tariq et al., 2023; Gao et al., 2023). Together, these factors control the contaminant mobility, redistribution, and persistence across the river basins. Despite this recognition, most existing assessments do not explicitly link surface and subsurface processes with vegetation vulnerability, nor do they identify which landscape zones are most susceptible and require prioritized monitoring and intervention (Ma et al., 2026).

Methodologically, earlier research has focused on quantifying heavy metal concentrations in soil and water to delineate contamination zones (Chen et al., 2025). While such approaches provide valuable information on pollution status, the combined influence of environmental drivers on vegetation response has remained insufficiently explored. Conventional ecological risk assessment frameworks assume vegetation as a uniform ecological receptor with similar exposure and response across vegetative organs (Zhao et al., 2025; Adams and Garman, 2024). In reality, vegetation responses to heavy metal contamination are strongly dependent on root depth and exposure pathways. Shallow-root vegetation (e.g., annual crops) is primarily influenced by heavy metals that are migrated through surface runoff, irrigation return flows, and topsoil deposition (Romeh et al., 2025). In contrast, deep-root vegetation (e.g., trees and perennial shrubs) is exposed to contaminants by vertical percolation and shallow groundwater recharge processes (Lv et al., 2025). These contrasting exposure mechanisms necessitate the development of vegetation ecological risk using a source-pathway-receptor (SPR) conceptual approach. However, its application has rarely been extended to resolve root depth-specific vegetation risk (Yu et al., 2025). Similarly, crop growth and hydrological models, such as the Agricultural Production Systems sIMulator (APSIM), AquaCrop, and the Soil and Water Assessment Tool (SWAT), simulate vegetation responses to water and nutrient dynamics but do not account for the cumulative effect of multisource environmental parameters (Gavasso-Rita et al., 2024). The selection of a method for determining parameter weights is also a crucial step in ecological risk estimation. Method based on equal weighting neglects the relative contribution of parameters, while expert-based approaches introduce subjectivity and inconsistencies in spatially heterogeneous environments. Beyond these methodological uncertainties, evaluating vegetation responses remains challenging due to limited field observations and the indirect nature of vegetation indicators. These methodological gaps are more pronounced in agro-industrial basins where field monitoring is sparse and hydrological observations are poorly gauged. Similar challenges have been reported in mixed agricultural-industrial landscapes across North America, Europe, and Asia (Niroula et al., 2023).

To address these knowledge gaps, this study develops a vegetation-based ecological risk (VER) assessment framework to evaluate and map root depth-specific vulnerability to heavy metal contamination in a polluted agro-industrial river basin. The framework follows a multistep structure, beginning with sub-basin scale assessment of integrated environmental risk (IER) as an initial screening indicator of cumulative environmental stress. Subsequently, VER is quantified following the source-pathway-receptor conceptual approach by integrating contaminant source intensity (CSI), environmental pathway intensity (EPI), and vegetation receptor vulnerability (VRV) using the entropy weighting method (EWM). The resulting risk patterns are evaluated against observed vegetation response indicators to assess their consistency and ecological relevance. Accordingly, this study addresses the following research questions: (a)

What are the spatial and seasonal distributions of heavy metals in soil and water? (b) How do hydrological processes, landscape characteristics, and anthropogenic pressures control the regional environmental risk pattern? (c) How do contaminant sources, environmental pathways, and vegetation vulnerability interact to produce vegetation-based ecological risk? (d) Which sub-basins and functional zones exhibit vegetation ecological risk and necessitate targeted mitigation strategies for sustainable management?

A key strength of the proposed framework is its ability to distinguish shallow- and deep-root vegetation vulnerability to dynamic heavy metal exposure through the interaction of contaminant sources, environmental pathways, and vegetation receptors. Consequently, the framework serves as a scientifically proven decision support and early warning tool and can be readily applied to other agro-industrial river basins facing similar data limitations and management challenges. This study will also inform environmental management strategies aimed at balancing water and soil pollution control with ecological function restoration.

2. Methodological framework for vegetation-based ecological risk assessment

The whole study is conducted using a systematic, multi-stage methodological framework designed to quantify vegetation-based ecological risk under heavy metal contamination (Fig. 1). The overall workflow is organized into six steps, beginning with spatial zone delineation and data preparation, followed by sub-basin scale environmental risk screening, vegetation-based ecological risk formulation, spatiotemporal analysis, validation, and management recommendation.

In this study, vegetation-based ecological risk (VER) is conceptualized as the likelihood and magnitude of adverse ecological effects on vegetation arising from the combined influence of contaminant sources, environmental transport pathways, and receptor-specific vulnerability. In agro-industrial river basins, seasonal hydrological variability, landscape heterogeneity, and contrasting root depth govern the exposure of vegetation to surface and subsurface contamination (Liu et al., 2022). To explicitly capture these interacting processes, the present study adopts the source-pathway-receptor (SPR) conceptual approach to represent the causal linkages between hazard generation, transport mechanisms, and ecological response. Following the SPR structure shown in Fig. 2, we have restructured VER from three interacting components: (i) contamination source intensity (CSI); (ii) environmental pathway intensity (EPI); and (iii) vegetation receptor vulnerability (VRV). Nevertheless, the computation of these components required the integration of multi-source environmental datasets, including heavy metal concentrations collected in-situ and analyzed under laboratory conditions. The sampling and measurement of heavy metals in surface water and soil are detailed in Supplementary Material 1. Subsequently, all environmental parameters were normalized and aggregated within ecologically meaningful sub-basins delineated using a digital elevation model (DEM).

2.1 Basin-scale screening using integrated environmental risk

Before the quantification of vegetation-based ecological risk, an integrated environmental risk (IER) was assessed at the sub-basin scale to characterize the cumulative magnitude and spatial heterogeneity of environmental disturbance (Bartan et al., 2023). Thus, IER functions as an initial screening layer to capture the combined effects of multiple environmental stressors. Parameters representing anthropogenic pressure, land-use characteristics, soil and terrain properties, hydrological variables, and heavy-metal concentrations were integrated using the entropy weighting method (EWM) to derive the IER. The resulting IER map identifies zones with dominant environmental effects and provides insight into the key drivers influencing the ecological security.

2.2 Quantification of source contamination intensity, pathway forcing, and receptor vulnerability

CSI defines the severity of heavy metal contamination from industrial effluents, wastewater discharges, agricultural runoff, and polluted irrigation inputs. These sources were divided into surface and subsurface hazards depending on their presence in the environment. EPI describes the surface and subsurface processes by which contaminants are mobilized and transported from sources to different vegetation root depths. VRV captures the sensitivity and adaptive capacity of vegetation to contaminant exposure. This incorporates vegetation characteristics such as rooting depth, physiological tolerance, land-cover conditions, and seasonal dynamics. The grouping of environmental parameters into the CSI, EPI, and VRV components was performed using multivariate analysis, as detailed in Supplementary Material 2.

2.3 Development of vegetation-based ecological risk

Following the risk assessment structure of the source-pathway-receptor approach, VER was computed by integrating CSI, EPI, and VRV. The general formulation of VER is expressed in Eq. 1. At the sub-basin scale (i), VER was computed as a weighted linear combination of the three components using weighted overlay technique in ArcGIS 10.7.1 as shown in Eq. 2. Here, f_{ij} represents the standardized value of each risk component for i sub-basin (Eq. 3) and, W_j denotes the corresponding weight derived using EWM (Chen et al., 2023). The detailed mathematical formula and procedure of the EWM are provided in Supplementary Material 3.

$VER = f(SCI, EPI, VRV) \quad (1)$
$VER_i = \sum_{j=1}^m f_{ij} \cdot W_j \quad (2)$
$f_{ij} = \begin{bmatrix} SCI_i, j = 1 \\ EPI_i, j = 2 \\ VRV_i, j = 3 \end{bmatrix} \quad (3)$

To assess the spatial variability of VER, the derived weights of each component were overlaid with root depth-specific vegetation land cover data. Accordingly, S-VER was formulated using surface CSI, surface EPI, and shallow root VRV, while D-VER was computed using subsurface CSI, subsurface EPI, and deep root VRV. Both S-VER and D-VER were also calculated for pre-monsoon and post-monsoon seasons to capture the seasonal effect of environmental pathways and contaminant mobility. The resulting risk maps distinguish surface and subsurface-driven vegetation risk across the region.

2.4 Validation of vegetation-based ecological risk

Heavy metal exposure develops physiological changes that influence the absorption and reflectance characteristics of vegetation leaves in the visible and near infrared region of the electromagnetic spectrum. Such spectral responses are effectively detected and quantified using remotely sensed satellite imagery. Among various vegetation indices, the normalized difference vegetation index (NDVI) is a widely accepted indicator for characterizing vegetation response due to its sensitivity to near infrared (NIR) and red (RED) reflectance bands. Previous studies by Mirzaei et al. (2019) and Li et al. (2015) reported that heavy metal accumulation alters vegetation spectral signatures, thereby reducing NDVI values in the vegetation. In the present study, validation of root depth-specific VER was performed using the NDVI at sub-basin (i) scale, which was computed following Eq. 4.

$$NDVI_i = \frac{(NIR - RED)_i}{(NIR + RED)_i}$$

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Furthermore, the relationship between seasonal S-VER and D-VER values and observed vegetation response was evaluated using Spearman's rank correlation analysis. Areas characterized by high VER and low NDVI were interpreted as zones where estimated risk aligns with actual vegetation stress.

3. Application

The developed framework of vegetation-based ecological risk assessment was applied to the Hindon River Basin (HRB) located in western Uttar Pradesh, Northern India (Fig. 3). The HRB, a sub-basin covering a catchment area of 5870 km² of the Yamuna River system, extends between latitudes 28°20'-29°30' N and longitudes 77°20'-78°20' E. The Hindon River flows southward for 394 km before joining the Yamuna River near Gautam Buddha Nagar. This river is hydrologically supported by major tributaries such as the Kali and Krishni rivers, along with minor tributaries including the Dhamola and Nagdehi.

Over the past two decades, HRB has witnessed a substantial loss of aquatic and human lives through heavy metal contamination in water and soil (Golui et al., 2022). Concurrently, rapid socio-economic development has profoundly altered the dynamics of regional natural resources. Agricultural expansion now dominates the landscape at the expense of forest cover, with nearly 68% of the basin area under water-intensive cash crops (Singh et al., 2024). This land-use transformation has resulted in excessive fertilizer and pesticide application, irrigation demand, large-scale surface water diversion, and intensive groundwater abstraction. In parallel, the agricultural growth has supported the proliferation of small- and large-scale industries, including sugar mills, distilleries, paper mills, tanneries, and electroplating industries (Gupta et al., 2023). Inadequate enforcement of wastewater treatment regulations has led to widespread illegal practices. The release of untreated effluents through concealed pipelines into canals, drains, and river stretches is one such persistent issue. Collectively, these activities have contributed to ecological and human health risks, manifested through declining crop productivity, fish mortality, and an increasing incidence of carcinogenic health disorders (Jadav and Yadav, 2025). Hydrologically, the HRB is experiencing a challenge of water availability driven by declining rainfall, rising temperatures, and sustained surface and groundwater withdrawals. The drying of the river stretches, and the depleting groundwater level are eye-witnessed examples in the recent site survey. The regional hydrology is further controlled by terrain slope and the diversion of water from the Upper Ganga and the Eastern Yamuna canal network (Dwivedi and Yadav, 2025). Considering the spatial heterogeneity in land use, socio-economic functions, and environmental pressure, a scientifically robust identification of vegetation ecological risk zones is required. Accordingly, the HRB was classified into major function-oriented zones (MFOZs), namely the Economic Development Zone (EDZ), Agricultural Production Zone (APZ), and Ecological Function Zone (EFZ). In addition, the HRB was subdivided into Upper (UHB), Middle (MHB), and Lower (LHB) regions based on terrain characteristics to capture downstream variations in hydrological processes, contaminant transport, and accumulation behavior.

3.1 Parameter identification, data collection, and preparation

To achieve the objectives of this study, a comprehensive suite of 36 environmental parameters was identified, collected, and processed for HRB (Table 1). These parameters were selected to represent contaminant sources, environmental transport pathways, and vegetation receptor characteristics, in accordance with the SPR risk assessment structure in the VER framework.

Table 1
Details of environmental parameters used in the present study.

Data type	Dataset description	Temporal / Spatial resolution	Data sources
Environmental pollution	Soil and surface water quality (Cr, Ni, Pb, Hg, As, Cd)	Pre-monsoon and post-monsoon (2023)	Field sampling and Laboratory analysis using Atomic Absorption Spectroscopy (AAS).
Landscape and terrain characteristics	Land use and cover (Built-up, water bodies, barren, forest, Kharif, Rabi, Zaid, etc.)	56 m (2023)	National Remote Sensing Centre (NRSC), Hyderabad, India.
	Slope class (0–2%, 2–4%, > 4%)	30 m (2023)	Shuttle Radar Topographic Mission (SRTM), USGS Earth Explorer.
	Soil texture (Clay, Sand, Silt)	1:500000 scale	Food and Agriculture Organization (FAO) global soil.
Hydrological variables	Surface runoff, evapotranspiration, percolation, water yield	Pre-monsoon and post-monsoon (2023)	Hydrological model (Dwivedi and Yadav, 2025).
Anthropogenic pressure indicators	Industrial density, distance to industry, irrigation intensity	Pre-monsoon and post-monsoon (2023)	Central Pollution Control Board and the district-level agricultural reports.
Vegetation response indicator	Normalized difference vegetation index	30 m (Pre-monsoon and post-monsoon 2023)	Landsat 8–9, USGS Earth Explorer.

4. Results

4.1 Statistical characterization of environmental parameters

Descriptive statistics of 36 environmental parameters used in this study are summarized in Table S1. Among the environmental pollution parameters, the elevated mean concentrations of heavy metals in surface water during the pre-monsoon season follow the order $Cr > Pb > Ni > Cd > Hg > As$. With reference to water quality guidelines prescribed by the Bureau of Indian Standards (BIS, 2012), we found 71.43% of the sub-basins (4069.88 km²) were impacted by Cr, while 65.71% (4002.71 km²) and 62.86% (3487.01 km²) were affected by Pb and Cd, respectively. Following the monsoon, mean metal concentrations declined substantially and followed the order $Cr > Ni > Pb > As > Cd > Hg$. Correspondingly, the proportion of affected sub-basins decreased to 2.86% (211.25 km²) for Cr, 11.43% (592.45 km²) for Ni, and 14.29% (736.03 km²) for Hg. In contrast to surface water contamination, soil heavy metal concentrations (Fig. 4) exhibited a consistent dominating pattern of $Cr > Ni > Pb > As > Cd > Hg$ across both seasons. Using the agricultural soil quality guidelines proposed by Kumar et al. (2019), we found nearly all sub-basins (5818.06 km²) to be unsafe for Cr, Ni, Cd, and Pb during the pre-monsoon season. In contrast, 22.86% (1526.82 km²) and 20% (1281.23 km²) of sub-basins were affected by As and Hg. A similar contamination pattern was recorded during the post-monsoon season, with Cr, Ni, Cd, and Pb consistently impacted all sub-basins, while only 8.57% (358.27 km²) and 5.71% (357.17 km²) of sub-basins were unsafe for As and Hg. The observed spatial distribution of each heavy metal concentration in surface water and soil during pre- and post-monsoon season are shown in Fig. S3 (a,b). Further analysis of coefficients of variation (CV) in heavy metals confirmed significant spatial heterogeneity rather than uniform distribution. In surface water, we observed extremely high CV values ($CV > 1$) for Ni, Cd, and Hg during the pre-monsoon season. However, soil heavy metals exhibited moderate CV in the range of 0.24–0.57. Interestingly, Hg displayed exceptionally high spatial variability with $CV > 3$ in both soil and water media during the pre-monsoon season. The regional scale analysis of all heavy metals and the influence of environmental parameters on their spatial variability are detailed in Supplementary Material 4.

4.2 Integrated environmental risk assessment

Figure 5 (a,b) shows the spatial distribution of IER across the HRB during pre- and post-monsoon seasons. Overall, IER levels exhibited a declining trend from pre-monsoon to post-monsoon season. However, despite this reduction, the areal extent of moderate and high-level IER increased during the post-monsoon period, with a corresponding decrease in low-level areas. The proportion of the sub-basin classified as low-level declined from 60% during the pre-monsoon to 43% in the post-monsoon season, corresponding to a cumulative reduction of 982.87 km². Conversely, the sub-basin extent of moderate- and high-level IER increased substantially, rising from 31% to 46% and from 3% to 11%, respectively, with cumulative increases of 807.76 km² and 175.11 km². Seasonally, IER levels exhibited pronounced variability across 60% of sub-basins. This variability was spatially heterogeneous, with significant seasonal transitions observed in 75%, 77%, and 20% of sub-basins within the upper, middle, and lower regions, respectively (Fig. 6a). An analysis of IER distribution across the regionally subdivided HRB revealed a clear seasonal transition from low-level to moderate- and high-level IER. Low-level IER remained dominant in 92% (1985.65 km²) and 83% (1908.02 km²) of the sub-basins of the UHB during the pre- and post-monsoon seasons. Only 9% (77.63 km²) of the low-level sub-basin area transitioned to moderate-level IER during the seasonal shift. In contrast, the MHB exhibited a pronounced decline in low-level IER from 54% (1418.46 km²) during the pre-monsoon season to 15% (513.22 km²) during the post-monsoon season. This shift was driven by strong seasonal effects, resulting in increases of moderate- and high-level IER by 24% and 15%, corresponding to conversions of 672.18 km² and 233.10 km² of low-level sub-basin area, respectively. The LHB showed comparatively limited seasonal variability, with low-level IER remaining constant at 30% (178.52 km²) of sub-basins in both seasons, primarily due to 10% (57.96 km²) of the sub-basins transitioning from high- to moderate-level IER. Across the MFOZs, low-level IER dominated in 63% (944.94 km²), 55% (1442.72 km²), and 83% (1194.97 km²) of sub-basins within the EDZ, APZ, and EFZ during the pre-monsoon season (Fig. 5c,d). The number of sub-basins contributing to seasonally varied IER levels across the different HRB regions and MFOZs is shown in Fig. 8(a). Under post-monsoon conditions, moderate IER became dominant in the EDZ and APZ, covering 63% (1291.02 km²) and 50% (1356.23 km²) of the total sub-basin area. In contrast, low-level IER remained prevalent within the EFZ, occupying 67% of the sub-basins.

4.3 Contamination source intensity

CSI was derived by integrating surface hazard (SH) and subsurface hazard (SSH) indicators. The result found moderate-level SH in 46% (2407.10 km²) and low-level SH in 80% (4534.45 km²) of the sub-basin during the pre- and post-monsoon season (Fig. S4). The observed seasonal shift resulted from the conversion of moderate (1852.39 km²) and high (595.85 km²) level sub-basin area into the low-level condition. Similarly, we observed low-level SSH in 57% (3049.02 km²) and moderate-level SSH in 51% (3332.22 km²) of the sub-basin, which were dominated during the pre- and post-monsoon season. This transition was primarily due to the reduction of 616.61 km² in the low-level SSH sub-basins to the moderate-level condition.

Across the HRB, 82.86% and 40% of sub-basins experienced dominant seasonal effects (Change < 0) in SH and SSH (Fig. 6b). This variability was consistently observed at the regional scale, with 83.33% (SH) and 75% (SSH) of sub-basins in UHB, 92% (SH) and 7.70% (SSH) sub-basins in MHB, and 70% (SH) and 40% (SSH) sub-basins in LHB, undergoing significant seasonal shifts. The dominant seasonal effect developed spatial heterogeneity in the distribution of SH and SSH across different regions of HRB. Moderate-level SH in 60% (699.76 km²) and low-level SSH in 80% (743.81 km²) of the sub-basin were dominated during the pre-monsoon season. These conditions were quickly transitioned in post-monsoon with SH showing strong low-level areal extent in 83% (2000.54 km²), 92% (1875.72 km²), and 60% (658.18 km²) of the sub-basins of UHB, MHB, and LHB. Similarly, SSH was also found to be dominated by low levels in 50% (976.46 km²) of the sub-basin of LHB.

Across the MFOZs, EDZ showed high-level SH in 50% (1021.01 km²) of sub-basins and low-level SH in 63% (1075.33 km²) during the pre- and post-monsoon season. On the other hand, APZ (50% (1110.41 km²) and 90% (2168.40 km²) of the sub-basins in pre- and post-monsoon seasons) and EFZ (50% (751.52 km²) and 67% (986.97 km²) of the sub-basins in pre- and post-monsoon seasons) were dominated by low-level SH across both seasons. Similar to SH, we analyzed the distribution of SSH across the MFOZs. The result highlighted a dominant low-level SSH distribution in EDZ (63% (1075.33 km²) of sub-basins in both seasons) and APZ (60% (1406.335 km²) and 50% (1149.10 km²) of the sub-basin in pre- and post-monsoon seasons). Nevertheless, EFZ witnessed a significant moderate level of SSH in 67% (986.98 km²) and 83% (1042.60 km²) of the sub-basin in pre- and post-monsoon seasons. The number of sub-basins contributing to each seasonally varied SH and SSH level across the different HRB regions and MFOZs is shown in Fig. 8(b).

4.4 Environmental pathway intensity

EPI was derived by combining surface (SHF) and subsurface hydrological forcing (SSHF) indicators. The result showed moderate-level SHF in 40% (2214.45 km²) and low-level SHF in 51% (2670.74 km²) of the sub-basin during pre- and post-monsoon seasons (Fig. S5). The observed change was attributed to a reduction of cumulative area of 980.88 km² in high-level SHF during the post-monsoon. Similarly, we observed moderate-level SSHF in 46% (2604.17 km²) and high-level SSHF in 46% (2261.99 km²) of the sub-basin during pre- and post-monsoon seasons. Figure 6(c) shows that the seasonal shift in SSHF causes a cumulative increase of 965.33 km² across 559.71 km² and 405.62 km² sub-basin areas under low- and moderate-levels. On analyzing the seasonal change across the HRB, we found 34.29% of sub-basins exhibiting intense seasonal effect (Change < 0) in SSHF, while only 8.57% of sub-basins exhibited seasonal shifting in SHF.

Regionally, MHB was observed with dominant moderate-level SHF in 69% (1402.80 km²) of sub-basins during the pre-monsoon, while LHB was observed with dominant low-level SHF in 100% (1472.71 km²) of sub-basins during the post-monsoon season. Notably, during the post-monsoon season, moderate-level SHF in 83% (1906.24 km²) of sub-basins was dominant in UHB. Similarly, we also observe a high level of SHF in LHB during the pre-monsoon season, covering 60% (373.43 km²) of the area. Similar analysis was continued in SSHF, where a significant moderate-level was reported in 50% (1013.57 km²) and 80% (1174.21 km²) of sub-basin of UHB and LHB during the pre-monsoon season. In contrast, MHB was found to exhibit high- to moderate-level SSHF in 46% (1088.67 km²) and 62% (1065.64 km²) of sub-basins during the pre- and post-monsoon seasons. Interestingly, 90% (1232.17 km²) of the sub-basin in LHB also showed a high-level of SSHF during the post-monsoon season.

Across the MFOZs, EDZ predominantly hold moderate- to high-level SHF in 50% (818.35 km²) and 25% (292.10 km²) of sub-basins during pre- and post-monsoon season. Similarly, APZ was observed with a strong high- to moderate-level SHF in 45% (1188.11 km²) and 45% (998.21 km²) of the sub-basins during the pre-monsoon season. In contrast, EFZ exhibited a dominant moderate-level of SHF in 83% (1194.97 km²) of the sub-basin during the post-monsoon season. On extending a similar analysis in SSHF, we observed EDZ under consistent high-levels of SSHF in 50% (773.36 km²) and 50% (1053.92 km²) during the pre- and post-monsoon season. In APZ, moderate-level SSHF strongly covered 60% (1271.88 km²) and 45% (1203.44 km²) of the sub-basin during pre- and post-monsoon seasons. Conversely, EFZ was reportedly seen with low-level SSHF in 67% (986.98 km²) and 17% (228.96 km²) of the sub-basin during pre- and post-monsoon seasons. The number of sub-basins contributing to seasonally varied SHF and SSHF levels across the different HRB regions and MFOZs is shown in Fig. 8(c).

4.5 Vegetation receptor vulnerability

VRV was estimated by integrating root depth-specific (shallow versus deep root) and seasonal (pre-monsoon versus post-monsoon) vegetation vulnerability. On analyzing the root depth-specific vegetation across the HRB, we found a moderate-level of shallow-root and a low-level of deep-root dominated in 77% (4847.67 km²) and 66% (3564.76 km²) of the sub-basin (Fig. S6). In contrast, only 6% and 17% of sub-basins were observed with high shallow- and deep-root vegetation. Regionally, shallow-root vegetation of moderate-level was observed in 100% (2055.96 km²), 67% (1576.46 km²), and 60% (1215.26 km²) of sub-basin of MHB, UHB, and LHB, respectively. In contrast, shallow-root vegetation of high-level was also noted in 20% (143.88 km²) of the sub-basin of LHB. Similar analysis was extended to deep-root vegetation, where high-levels were recorded in 33% (828.05 km²) and 20% (117.73 km²) of sub-basins in UHB and LHB. Nevertheless, a low-level of deep-root vegetation was also recorded in 42% (765.39 km²) and 77% (1444.38 km²) of the sub-basins of UHB and MHB. Across the MFOZs, shallow-root vegetation of moderate-level was reported in 90% (2234.75 km²) and 75% (1746.28 km²) of sub-basins within APZ and EDZ.

However, a low-level of shallow root vegetation was observed in 67% of sub-basins within EFZ. Similar analysis was continued in deep root vegetation, where we observed that EFZ alone exhibited dominant high-level deep-root vegetation in 50% (499.07 km²) of sub-basins. However, APZ and EDZ were noted with low-levels of deep-root vegetation in 85% (2000.87 km²) and 63% (71.93 km²) of sub-basins. The number of sub-basins contributing to seasonally varied shallow- and deep-root vegetation levels across the different HRB regions and MFOZs is shown in Fig. 8(d).

Further, we analyzed the seasonal distribution of vegetation across the HRB. The result suggested that vegetation during the pre-monsoon season cover 66% (3564.76 km²) of sub-basins, while the post-monsoon season cover 54% (2899.07 km²) of sub-basins across the HRB (Fig. 6d). As a whole, 45.71% of sub-basins underwent a change in vegetation cover from pre- to post-monsoon season. These seasonal changes were regionally dominant in LHB (80% of sub-basins) and MHB (46.15% of sub-basins). Regionally, low-level vegetation cover dominated the MHB and LHB during both the pre-monsoon (77% (1444.38 km²) and 80% (1354.98 km²) of sub-basins) and post-monsoon (85% (1681.47 km²) and 30% (452.20 km²) of sub-basins) seasons. In contrast, high-levels of pre-monsoon (33% and 20% of sub-basins) and post-monsoon (25% and 60% of sub-basins) vegetation were reported in UHB and LHB. Across the MFOZs, low-levels of pre- and post-monsoon vegetation were observed in 85% (70%) and 63% (50%) of sub-basins in APZ and EDZ, respectively. However, EFZ was covered with high-levels of pre- and post-monsoon vegetation in 50% of the sub-basins.

4.6 Spatiotemporal patterns of shallow and deep root vegetation-based ecological risk

We observed a dominant moderate-level S-VER in 49% (2437.10 km²) and 43% (2762.05 km²) of the sub-basin during the pre- and post-monsoon seasons. Moreover, sub-basins under high-level (610.22 km²) were reportedly transitioned to low (285.27 km²) and moderate-level (324.95 km²) after the monsoon season (Table 2). The observed transition was due to the seasonal shift (change < 0) in 94.28% of sub-basins across the HRB. A similar pattern remained regionally consistent across 100% of sub-basins in the UHB, 92.31% in the MHB, and 90% in the LHB (Fig. 6e).

We further analyzed the S-VER distribution regionally, and the results suggested that LHB exhibits dominant moderate-level in 70% (729.76 km²) and 40% (872.78 km²) of the sub-basin during pre- and post-monsoon seasons (Fig. 7). Moreover, 20% of sub-basins under pre-monsoon high-level S-VER within LHB were completely changed to low-levels during the post-monsoon season. In contrast, MHB and UHB were observed with a strong low- (46% in pre-monsoon and 54% in post-monsoon) and moderate-level (33% in pre-monsoon and 67% in post-monsoon) S-VER, respectively. Across the MFOZs, moderate- and low-level S-VER were dominant in the APZ and EFZ, whereas moderate- and high-level S-VER were dominant in the EDZ. In APZ, moderate- and low-level S-VER were represented by 55% (1298.22 km²) and 45% (1080.41 km²) of sub-basins during the pre-monsoon season, and 35% (785.86 km²) and 60% (1440.36 km²) of sub-basins during the post-monsoon season. Similarly, in EFZ, moderate- and low-level S-VER were observed in 50% (499.07 km²) and 50% (751.52 km²) of the sub-basins during the pre-monsoon season, and 50% (651.44 km²) and 50% (599.15 km²) of the sub-basins during the post-monsoon season. In EDZ, moderate- and high-level S-VER were noted in 38% (639.81 km²) and 50% (1021.01 km²) of the sub-basins during the pre-monsoon season, and 50% (1021.01 km²) and 38% (562.11 km²) of the sub-basins during the post-monsoon season. The number of sub-basins contributing to seasonally varied S-VER and D-VER levels across the different HRB regions and MFOZs is shown in Fig. 8(e).

Similar analyses were conducted in D-VER, and results indicated dominant low- (51% (2689.66 km²) and 46% (2432.41 km²) of sub-basins) to moderate- (46% (3074.98 km²) and 51% (3332.22 km²) of sub-basins) levels of D-VER during the pre- and post-monsoon seasons (Fig. 7d). Moreover, 5% (257.24 km²) sub-basins of low-level D-VER was found to be transitioned to moderate-levels as a result of dominant seasonal change (change < 0) in 65.71% of sub-basins of the HRB (Fig. 6e). These seasonal changes were regionally reported in 100%, 23.08%, and 80% of sub-basins of UHB, MHB, and LHB, respectively. We further analyzed the regional distribution of the D-VER pattern across the HRB. The result suggested strong low-level D-VER distributed in 70% (688.18 km²) and 50% (496.25 km²) of the sub-basins of LHB during the pre- and post-monsoon seasons. Similarly, moderate-level D-VER was reported in 58% (1602.32 km²) and 75% (1807.39 km²) of the sub-basins of UHB. In contrast, MHB exhibited only 8% sub-basins with a high-level of D-VER in both seasons. Across MFOZs, EDZ and APZ were found with strong low-level D-VER in 63% (1075.33 km²) and 50% (1149.10 km²) of the sub-basins during both seasons. However, EFZ exhibited a significant moderate level of D-VER in 83% (1042.60 km²) of the sub-basins during both seasons.

4.7 Validation of shallow and deep-rooted vegetation-based ecological risk

Figure 9 shows the Spearman's rank correlation between estimated root depth-specific vegetation risk and vegetation response from NDVI. The result revealed a negative correlation of $\rho = -0.29$ between S-VER and S-NDVI during the pre-monsoon season. This correlation was found to be stronger during the post-monsoon season with $\rho = -0.36$. We further examined the cross-seasonal correlation between S-VER (pre) and S-NDVI (post), and yet the result exhibited a negative correlation with $\rho = -0.20$. In contrast, D-VER showed high buffering capacity during the pre-monsoon season, with weak to negligible correlation observed with NDVI ($\rho = -0.11$). In contrast, the post-monsoon season improved vegetation greenness, and therefore, a moderate positive association of $\rho = 0.76$ was found with D-NDVI. On examining the correlation of D-VER across the season, we observed high temporal stability with a correlation ranging between 0.83 and 0.91.

5. Discussion

This study reveals the distinct exposure mechanisms for ecological risk in shallow- and deep-root vegetation through interaction of contamination source, environmental pathways, and vegetation receptor characteristics across the HRB. The contrasting spatiotemporal variability in S-VER and D-VER shows that the vegetation response to contamination is depth-dependent and seasonally asymmetric. Shallow-root vegetation is sensitive to strong seasonal variability driven by surface contamination dynamics and hydrological forcing. Conversely, deep-root vegetation responds to persistent ecological stress driven by long-term subsurface metal accumulation. These patterns are further supported by declining vegetation greenness indicated by lower NDVI values within high S-VER zones, whereas deep-root vegetation exhibits delayed and buffered spectral responses.

Table 2

Regional distribution of shallow-root vegetation-based ecological risk (S-VER) and deep-root vegetation-based ecological risk (D-VER) across the HRB.

Region	Classification	Shallow-root vegetation-based ecological risk (S-VER)					Deep-root vegetation-based ecological risk (D-VER)				
		Percentage of Sub-basin (Pre-monsoon)	Percentage of Sub-basin (Post-monsoon)	Area in Km ² (Pre-monsoon)	Area in Km ² (Post-monsoon)	Change in Area (km ²)	Percentage of Sub-basin (Pre-monsoon)	Percentage of Sub-basin (Post-monsoon)	Area in Km ² (Pre-monsoon)	Area in Km ² (Post-monsoon)	Change in Area (km ²)
HRB	Low	37.00	46.00	2056.21	2341.48	-285.27	51.00	46.00	2689.66	2432.41	-257.25
	Moderate	49.00	43.00	2437.10	2762.05	-324.95	46.00	51.00	3074.98	3332.22	-257.24
	High	14.00	11.00	1324.75	714.53	610.22	3.00	3.00	53.42	53.42	0.00
UHB	Low	50.00	25.00	1096.26	774.27	321.99	42.00	25.00	687.08	482.01	205.07
	Moderate	33.00	67.00	808.53	1362.71	-554.18	58.00	75.00	1602.32	1807.39	-205.07
	High	17.00	8.00	384.60	152.42	232.19	0.00	0.00	0.00	0.00	0.00
MHB	Low	46.00	54.00	945.90	967.29	-21.39	46.00	62.00	1314.40	1454.16	-139.76
	Moderate	46.00	23.00	898.81	526.56	372.25	46.00	31.00	688.14	548.37	139.77
	High	8.00	23.00	211.24	562.11	-350.87	8.00	8.00	53.42	53.42	0.00
LHB	Low	10.00	60.00	14.05	599.93	-585.88	70.00	50.00	688.18	496.25	191.93
	Moderate	70.00	40.00	729.76	872.78	-143.03	30.00	50.00	784.53	976.46	-191.93
	High	20.00	0.00	728.90	0.00	728.90	0.00	0.00	0.00	0.00	0.00

5.1 Influence of contamination sources and landscape characteristics on VER

The integrated environmental risk identifies hydrological connectivity and land-use intensity as the dominant basin-scale drivers controlling the spatial redistribution of environmental stress and the resulting vegetation ecological risks (Fig. 10e). In practice, this influence is reflected across the HRB, where expanding agricultural activities for water-intensive cash crops such as sugarcane, wheat, rice, mustard, and edible vegetables have promoted the proliferation of sugar mills, distilleries, and agro-processing industries. Effluent discharges from these industries, together with domestic wastewater and agricultural return flows, have resulted in increased heavy metal loading in rivers, canals, and soils (Sharma et al., 2021). Moreover, the use of contaminated surface water for irrigation, together with return flows from the topsoil layer, develops a recurring source-sink contamination cycle. This mechanism strongly exposes shallow-root vegetation to Hg, Cd, and Pb originating from built-up, agriculture, and nearby industrial areas (Fig. 10a-i). Soils enriched in clay and silt further intensify this exposure by promoting surface adsorption and retention of metals (Xie et al., 2024; Selim et al., 2023). Hu et al. (2023) also reported high heavy metal accumulation in peri-urban soils and cultivated vegetables, which causes potential human health risks. Consequently, elevated S-VER is predominantly observed in intensely cultivated sub-basins within the MHB and LHB (Fig. 10d-i). Within these regions, rabi and zaid, including double- and triple-crop experience the highest levels of exposure (Fig. 10c-i). In contrast, sub-basins dominated by coarser soil texture facilitate the downward leaching of Hg, Cd, and As into deeper subsurface layers (Fig. 10a-ii). This persistent subsurface contamination contributes to relatively stable low-to-moderate D-VER patterns observed under plantation and forest cover across the HRB, as shown in Fig. 10c-ii and Fig. 10d-ii.

5.2 Influence of hydrological pathways in regulating metal mobility

Seasonal monsoonal rainfall and localized contamination inputs from concealed and open drains alter surface runoff, water yield, evapotranspiration, and percolation processes, thereby modifying heavy metal mobility and exposure pathways. In 2023, water availability across the HRB was higher during the pre-monsoon season compared to the post-monsoon season (Dwivedi and Yadav, 2025; Jadav and Yadav, 2025). High surface runoff and river flows periodically reduce heavy metal concentrations through the dilution process; however, they also lead to expansion of contaminated zones via downstream redistribution depending on terrain slope (Aziz et al., 2023) (Fig. 10b-i). Intensive irrigation practices using contaminated surface water further enhance the wet-dry cycle, enabling the accumulation and migration of heavy metals (Tudi et al., 2025). Under these conditions, MHB with flatter terrain favours metal accumulation, whereas steeper reaches of LHB facilitate rapid transport and convergence of upstream contaminants. These coupled processes explain the observed seasonal transition from high to moderate S-VER during the post-monsoon season (Li et al., 2025). At the subsurface level, percolation and water yield exert dominant control over vertically downward metal movement in both seasons, while evapotranspiration plays a comparatively minor role (Fig. 10b-ii). These processes enhance the exposure of deep-root vegetation to long-term storage within the soil profile (Hu et al., 2021). Unlike surface pathways, subsurface contamination remains buffered from short-term hydrological fluctuations as indicated by relatively stable D-VER patterns across seasons.

5.3 Spatiotemporal behaviour of VER across regions and functional zones

The UHB exhibited strong seasonal variability in both S-VER and D-VER due to dominant hydrological forcing and comparatively low contamination intensity. In contrast, the MHB serves as a transition zone, where mixed agricultural and built-up areas interact with moderate hydrological forcing, resulting in heterogeneous risk patterns within EDZ and APZ. The LHB has been consistently observed as a long-term ecological risk hotspot due to cumulative upstream contaminant migration, intensified surface hydrological forcing, and reduced dilution capacity. As a result, this region observed high to moderate S-VER and persistent D-VER across seasons. Across MFOZs, EDZ and APZ were dominated by shallow-root vegetation risk due to frequent irrigation and

surface exposure. In contrast, EFZ within UHB exhibited long-term soil metal accumulation in deep-root vegetation due to subsurface contamination migration processes (Deng et al., 2025).

5.4 Implications for VER management

The identification of root depth-specific ecological risk zones provides scientific guidance for targeted management strategies. Sub-basins exhibiting high S-VER should be regulated for irrigation water quality, interception of agricultural runoff pathways, and strict control of untreated wastewater discharge (Ferri and Bolelli, 2025). Areas dominated by D-VER require interventions such as phytostabilization, vegetative barriers, and long-term monitoring of subsurface contamination (Calmon et al., 2026). Priority should be given to low-risk sub-basins to create ecological buffers through restricting industrial expansion and promoting sustainable agricultural practices. Importantly, downstream EDZ and APZ require integrated basin-scale management to prevent continuous accumulation and ecological intensification.

5.5 Limitations and future research

Despite the scientific contribution, this study has a few limitations. Seasonal mean concentrations may not capture short-term contamination pulses during extreme rainfall and episodic discharges. Sampling density was limited to one representative site per sub-basin, and vegetation exposure was inferred through root-zone contamination rather than direct plant tissue analysis. Future studies should incorporate multi-site sampling, plant tissue measurements, sub-basin level crop yield validation, high-frequency monitoring, and coupled reactive transport modelling to better quantify metal migration and long-term storage within the vadose zone. Integration with geophysical techniques such as electrical resistivity tomography could further strengthen subsurface risk characterization.

6. Conclusions

The assessment of seasonal ecological risk due to heavy metal dynamics in shallow- and deep- root vegetation proven to be an effective approach to identify priority conservation areas and strengthen ecological security. Previous studies had simply evaluated the ecological risk using soil and water quality parameters, while the implications of contaminant transfer to vegetation roots remained insufficiently addressed. In contrast, the present study systematically combined the SCI, EPI, and VRV within a SPR approach. Preliminary investigation through IER revealed that environmental risk was not uniform across the HRB but dynamically transitioned between low- and high-level under changing hydrological conditions. These findings underscore the necessity of root depth-specific ecological risk rather than uniform vegetation assessments. From pre- to post-monsoon season, a moderate-level S-VER in 49% (1324.75 km²) and 43% (2762.05 km²) of the sub-basins was concentrated in agriculturally and hydrologically connected LHB. However, UHB and MHB were largely dominated by low-risk conditions. The observed risk levels also underwent seasonal redistribution due to the strong sensitivity of shallow-root vegetation to hydrological variability in nearly 94.28% of sub-basins. In contrast, D-VER was dominated by low-risk conditions with 51% (2689.66 km²) and 46% (2432.41 km²) of the sub-basins during pre- and post-monsoon season. Although 65.71% of sub-basins exhibited seasonal transitions, the overall D-VER pattern remained spatially stable. Low-levels of D-VER were concentrated in the LHB, whereas moderate and high-level was observed in the UHB and MHB. Following monsoonal rainfall, the decline in high S-VER sub-basins and the expansion of low-risk conditions in 80% (4534.45 km²) of the sub-basins indicate the redistribution of surface contamination by surface hydrological forcing. Similarly, increased subsurface hazard, together with enhanced subsurface hydrological forcing during recharge periods, led to persistent moderate-level D-VER exposure in deep-rooted vegetation. The dominance of shallow-root vegetation in more than 77% (4847.67 km²) of the sub-basins within APZ indicates high susceptibility to surface contamination and elevated S-VER. Conversely, the higher proportion of deep-root vegetation in the EFZ led to prolonged subsurface exposure and persistent moderate D-VER conditions. Among the MFOZs, EDZ were primarily controlled by SH and SHF, whereas APZ exhibited a strong interaction between shallow root vegetation vulnerability and SHF. In contrast, EFZ were dominantly influenced by SSH and SSHF. The overall findings emphasize that root depth-specific seasonal ecological risk assessment should be adopted in place of a uniform vegetation risk assessment framework for targeted pollution control, adaptive agricultural practices, and long-term subsurface monitoring and restoration measures in agro-industrial river basins.

Declarations

Authorship contribution statement

Prabhat Dwivedi: Writing - original draft, Software, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; Brijesh Kumar Yadav: Writing - review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no competing financial or personal interests that could have influenced the research presented in this work.

Acknowledgement

The authors are grateful to the Prime Minister Research Fellowship, Department of Science and Technology, and the Dutch Research Council for funding field visits of this research work under the project titled "Co-Creating Sustainable Agri-Water Use in the Hindon Sub-Basin." The authors are also thankful to the Central Water Commission for discharge data, the India Meteorological Department for meteorological data, the National Remote Sensing Centre, and IIT Roorkee for laboratory facilities.

Data Availability

The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

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